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M. I. Freidlin
A. D. Wentzell

Random Perturbations of Dynamical Systems

Translated by Joseph Szücs

With 20 Illustrations



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Foreword

Asymptotical problems have always played an important role in probability theory. In classical probability theory dealing mainly with sequences of independent variables, theorems of the type of laws of large numbers, theorems of the type of the central limit theorem, and theorems on large deviations constitute a major part of all investigations. In recent years, when random processes have become the main subject of study, asymptotic investigations have continued to play a major role. We can say that in the theory of random processes such investigations play an even greater role than in classical probability theory, because it is apparently impossible to obtain simple exact formulas in problems connected with large classes of random processes.

Asymptotical investigations in the theory of random processes include results of the types of both the laws of large numbers and the central limit theorem and, in the past decade, theorems on large deviations. Of course, all these problems have acquired new aspects and new interpretations in the theory of random processes.

One of the important schemes leading to the study of various limit theorems for random processes is dynamical systems subject to the effect of random perturbations. Several theoretical and applied problems lead to this scheme. It is often natural to assume that, in one sense or another, the random perturbations are small compared to the deterministic constituents of the motion. The problem of studying small random perturbations of dynamical systems has been posed in the paper by Pontrjagin, Andronov, and Vitt [1]. The results obtained in this article relate to one-dimensional and partly two-dimensional dynamical systems and perturbations leading to diffusion processes. Other types of random perturbations may also be considered; in particular, those arising in connection with the averaging principle. Here the smallness of the effect of perturbations is ensured by the fact that they oscillate quickly.

The contents of the book consists of various asymptotic problems arising as the parameter characterizing the smallness of random perturbations converges to zero. Of course, the authors could not consider all conceivable schemes of small random perturbations of dynamical systems. In particular, the book does not consider at all dynamical systems generated by random

vector fields. Much attention is given to the study of the effect of perturbations on large time intervals. On such intervals small perturbations essentially influence the behavior of the system in general. In order to take account of this influence, we have to be able to estimate the probabilities of rare events, i.e., we need theorems on the asymptotics of probabilities of large deviations for random processes. The book studies these asymptotics and their applications to problems of the behavior of a random process on large time intervals, such as the problem of the limit behavior of the invariant measure, the problem of exit of a random process from a domain, and the problem of stability under random perturbations. Some of these problems have been formulated for a long time and others are comparatively new.

The problems being studied can be considered as problems of the asymptotic study of integrals in a function space, and the fundamental method used can be considered as an infinite-dimensional generalization of the well-known method of Laplace. These constructions are linked to contemporary research in asymptotic methods. In the cases where, as a result of the effect of perturbations, diffusion processes are obtained, we arrive at problems closely connected with elliptic and parabolic differential equations with a small parameter. Our investigations imply some new results concerning such equations. We are interested in these connections and as a rule include the corresponding formulations in terms of differential equations.

We would like to note that this book is being written when the theory of large deviations for random processes is just being created. There have been a series of achievements but there is still much to be done. Therefore, the book treats some topics that have not yet taken their final form (part of the material is presented in a survey form). At the same time, some new research is not reflected at all in the book. The authors attempted to minimize the deficiencies connected with this.

The book is written for mathematicians but can also be used by specialists of adjacent fields. The fact is that although the proofs use quite intricate mathematical constructions, the results admit a simple formulation as a rule.

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